

BUSINESS REQUIREMENTS DOCUMENT

Personal AI Subscription Management Platform

Centralized tool inventory, recurring and on-demand cost control, token usage, payment-source governance, budgets, renewals, and alerts

Document	Value
Version	1.0
Status	Baseline for stakeholder review and solution delivery
Prepared for	Personal AI Subscription Management initiative
Target technology	PHP Laravel, MySQL, Metronic UI template
Prepared on	29 July 2026
Classification	Internal / Confidential

PURPOSE Define the complete business, functional, UX, data, security, reporting, and acceptance requirements for a compact, mobile-responsive system that gives users reliable control of all AI and related software subscriptions.

Document Control

Version	Date	Owner	Change summary	Status
1.0	29 Jul 2026	Product Owner	Initial complete BRD baseline	For review

Approval Record

Role	Name	Decision	Date	Comments
Business Sponsor	TBD	Pending	TBD	
Product Owner	TBD	Pending	TBD	
Technical Lead	TBD	Pending	TBD	
QA Lead	TBD	Pending	TBD	

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DOCUMENT USE Business stakeholders approve scope and rules; UX teams design against screen requirements; engineering implements the numbered requirements; QA derives tests from use cases and the traceability matrix.

1. Executive Summary

Individuals and small teams increasingly pay for a mix of AI chat products, model APIs, token packages, developer utilities, design products, storage, automation platforms, and other digital tools. Charges may be monthly, yearly, prepaid, usage-based, trial-based, or hybrid. They may be billed in different currencies and to different cards or accounts. Without a single control point, renewals are missed, unused products remain active, token spending exceeds expectations, and the true monthly cost is difficult to understand.

The proposed Personal AI Subscription Management Platform will provide one source of truth for subscriptions, usage and payment responsibility. It will normalize recurring and on-demand costs into monthly and annual views, track planned targets against actual and forecast spend, warn users before thresholds or renewals are crossed, and preserve a complete cost and lifecycle history.

1.1 Proposed Business Capability

- Register and manage AI subscriptions, AI token/API plans, other software tools, and custom subscription types.
- Support free, trial, monthly, quarterly, annual, prepaid-credit, on-demand, and hybrid charging models.
- Track recurring fees, add-ons, taxes, discounts, token purchases, metered usage, credits, refunds, and manual adjustments.
- Manage masked payment accounts such as debit/credit cards, bank accounts, wallets, virtual cards, and vendor balances.
- Set global, category, tool, and on-demand targets with warning and critical thresholds.
- Forecast end-of-period spend and projected renewal obligations using configurable costing rules.
- Send in-app and email alerts for target crossings, renewal windows, payment expiry, trial expiry, unusual usage, and missing data.
- Provide compact dashboards, responsive workflows, accessible forms, reports, exports, and an auditable history.

1.2 Business Value

Outcome	Expected value
Visibility	A reliable monthly, annual, category, tool, and payment-source view of commitments and usage.
Cost control	Early warning before spend exceeds target and clear identification of unused or duplicative tools.
Renewal control	Reduced accidental renewal, expired trials, service interruption, and late cancellation.
Governance	Traceable ownership, payment responsibility, approval status, changes, and exported evidence.
Decision support	Comparable effective monthly costs, cost trends, utilization, and savings opportunities.

2. Business Context, Objectives and Success Measures

2.1 Problem Statement

Subscription records are commonly dispersed across inboxes, vendor portals, bank statements, spreadsheets, and memory. A charge amount alone does not show whether it is a monthly commitment, annual renewal, token top-up, or on-demand usage. This fragmentation makes budgeting reactive and creates avoidable financial and operational risk.

2.2 Objectives

ID	Objective	Business measure
OBJ-01	Create one complete inventory of active, trial, paused, cancelled, and expired tools.	At least 95% of known subscriptions recorded after onboarding.
OBJ-02	Show actual, committed, and forecast costs by period.	Monthly totals reconcile to source data within approved tolerance.
OBJ-03	Prevent surprise overages and renewals.	Threshold and renewal notifications sent within configured windows.
OBJ-04	Make cost ownership and payment source clear.	Every paid subscription linked to an owner and active payment account or documented exception.
OBJ-05	Support cost optimization decisions.	Dashboard identifies unused, low-use, duplicate, or high-growth subscriptions.
OBJ-06	Provide a fast, usable experience on desktop and mobile.	Core tasks complete without horizontal scrolling at supported mobile widths.

2.3 Key Performance Indicators

- Inventory coverage: registered paid and trial tools divided by known tools.
- Spend variance: actual month-to-date and forecast end-of-month versus target.
- Prevented renewal value: value of subscriptions cancelled before unwanted renewal.
- Reconciliation rate: cost entries matched to a subscription and payment account.
- Notification effectiveness: alerts delivered, acknowledged, snoozed, and acted upon.
- Data freshness: percentage of active usage-based subscriptions updated within their expected cycle.
- Task efficiency: median time to add a subscription, record a usage charge, and identify the next renewal.

2.4 Definition of Success

MINIMUM SUCCESS A user can register all subscription models, see normalized monthly cost and forecast, set targets, receive actionable alerts, identify the card/account used, and produce an accurate cost report from any supported device.

3. Scope, Stakeholders, Assumptions and Constraints

3.1 In Scope

- User authentication, profile, currency, locale, time-zone, notification, and dashboard preferences.
- Tool/vendor catalogue and user-defined categories, tags, ownership, purpose, environment, and status.
- Subscription type configuration and lifecycle management for recurring, prepaid, usage-based, and hybrid plans.
- AI token/API pricing, packages, credits, usage entries, budget targets, forecast, and effective-unit cost.
- Payment-account register with masked details, expiry, status, billing currency, owner, and linked subscriptions.
- Cost ledger, taxes, discounts, refunds, credits, adjustments, proration, exchange-rate capture, and attachments.
- Targets, threshold policies, alert delivery, renewal reminders, trial reminders, unusual-cost detection, and acknowledgement.
- Dashboards, calendars, searchable lists, comparison, cost trends, reports, CSV/XLSX export, and printable summaries.
- Administrative reference data, role permissions, audit history, background jobs, system health, backup, and retention.
- Responsive web interface built with Laravel, MySQL, and a licensed Metronic template package.

3.2 Out of Scope for the Initial Release

- Initiating or settling vendor payments, changing a vendor subscription directly, or storing full card numbers/CVV.
- Automatic scraping of arbitrary vendor portals without an approved API or user-provided import.
- Corporate procurement, purchase orders, complex multi-entity accounting, tax filing, or general-ledger posting.
- Native iOS/Android applications; the first release is a responsive web application and may later become a PWA.
- Real-time bank aggregation unless delivered as a separately approved integration.
- Vendor-specific token measurement where the vendor does not expose usage data; manual entry/import remains supported.

3.3 Stakeholders and Roles

Stakeholder / role	Primary interest	Responsibilities
Business Sponsor	Value, funding, delivery outcome	Approve scope, budget, priorities, and go-live.
Product Owner	Requirement quality and value	Own backlog, rules, acceptance, and stakeholder decisions.
Account Owner / User	Personal subscription control	Maintain tools, targets, payment sources, and actions.

Stakeholder / role	Primary interest	Responsibilities
Viewer / Auditor	Read-only evidence	Review dashboard, reports, costs, and history.
Administrator	Configuration and support	Manage reference data, users, roles, jobs, templates, and support actions.
Finance Reviewer (optional)	Cost accuracy	Review reconciliation, adjustments, reports, and exceptions.
Engineering Team	Feasible implementation	Design, build, migrate, operate, and document the platform.
QA / Security	Quality and control	Verify requirements, security, accessibility, performance, and recovery.

3.4 Assumptions

- The first deployment is single-tenant or personal-workspace oriented; data structures should permit future multi-user workspaces.
- The user chooses one base reporting currency; transaction and subscription currencies are retained separately.
- Exchange rates may be entered manually or obtained from an approved provider; the rate, source, and effective date are stored.
- Notification delivery initially includes in-app and email; optional browser push, SMS, or chat channels require separate credentials.
- A subscription may have multiple cost components and plan changes over time; financial history is immutable except by controlled adjustment.
- The system is a management and forecasting tool, not a legal, accounting, or payment-processing system.

3.5 Constraints and Dependencies

- Development stack: supported PHP and Laravel release selected at delivery start, MySQL 8-compatible database, Metronic assets, queue workers, scheduler, and email provider.
- Use of Metronic is subject to the purchaser holding the correct commercial license; no third-party credentials or paid assets are embedded in source control.
- Vendor APIs differ in availability and billing semantics; every automated connector must degrade safely to manual entry or file import.
- Payment account data must be masked and minimized. Full PAN, CVV, online-banking password, private API key, and secret token storage are prohibited.
- Final tax, privacy, retention, and accessibility obligations depend on the deployment jurisdiction and must be confirmed before production.

4. Business Concepts, Rules and Cost Calculations

4.1 Core Definitions

Term	Definition
Tool	An AI or non-AI software product whose access, use, or cost is managed.
Vendor	The party billing for a tool; one vendor may supply multiple tools.
Subscription	A time-bounded commercial arrangement for a tool, including free, trial, recurring, prepaid, on-demand, or hybrid models.
Plan / rate version	The price and entitlement configuration effective for a defined date range.
Cost entry	A ledger event such as recurring fee, usage charge, token purchase, tax, discount, credit, refund, or adjustment.
Token package	A purchased or granted balance of units/credits/tokens with optional expiry.
Payment account	A masked card, bank account, wallet, virtual card, or vendor balance used to pay a charge.
Target	An approved spending ceiling or reference amount for a period, scope, and currency.
Alert policy	Thresholds, time windows, channels, recipients, quiet hours, and repetition rules.
Committed cost	A known contractual cost expected in the selected period regardless of usage.
Forecast cost	Projected total cost for the period using actuals plus future commitments and estimated on-demand usage.

4.2 Subscription Type Model

Type	Required characteristics	Cost treatment
Free	No charge; optional feature/usage tracking	Zero committed cost.
Trial	Start, end, post-trial action, expected conversion price	Forecast may include post-trial fee if conversion is enabled.
Monthly recurring	Billing anchor, amount, renewal, proration rule	Full or prorated fee in each billing period.
Annual recurring	Renewal date, annual fee, optional installment plan	Cash view at renewal; normalized monthly view equals annual net cost / 12.
Other recurring	Interval unit and count, e.g., quarterly or every 4 weeks	Occurrences generated by recurrence schedule.
Prepaid token / credit	Package units, purchase cost, expiry, balance	Cash at purchase; consumption and effective-unit cost tracked separately.
On-demand	Meter, rate tiers, minimums, caps, usage source	Actual usage x applicable rate plus fixed/minimum charges.
Hybrid	Recurring base fee plus included units and overage tiers	Base commitment plus forecast/actual overage.
One-time	Purchase date and amount	Recognized once; optional amortized management view.

4.3 Business Rules

Rule ID	Rule
BR-001	Every subscription belongs to exactly one tool and has one active type at a time.
BR-002	Plan, price, currency, billing cadence, tax treatment, and included entitlements are versioned by effective date.

Rule ID	Rule
BR-003	A paid subscription must reference a payment account unless it is marked reimbursed, invoice/manual, vendor credit, or exception with reason.
BR-004	Payment account identifiers display only friendly name, issuer/provider, type, last four digits or alias, owner, expiry month/year, and status.
BR-005	Historical cost entries are not silently recalculated when a current price, target, or exchange rate changes.
BR-006	The system stores original amount/currency and base-currency amount/rate/source for each financial event.
BR-007	Refunds, credits, and reversals reference the original entry where possible and never delete the audit record.
BR-008	A target has a scope, period, amount, currency, warning threshold, critical threshold, and status.
BR-009	Threshold comparisons use the selected basis: actual, committed, forecast, or actual-plus-committed.
BR-010	A threshold event is deduplicated for the configured cool-down window but may escalate when a higher threshold is crossed.
BR-011	On-demand forecast uses a user-selected method: fixed target, run-rate, recent-period average, manual forecast, or zero until usage is recorded.
BR-012	Trial and renewal dates are evaluated in the workspace time zone, with reminders scheduled by calendar date rather than server time.
BR-013	Pausing stops future recurring forecast only when the vendor contract is also paused; local status alone does not erase commitments.
BR-014	Cancellation retains historical costs and may include an access-end date, cancellation deadline, refund, or cancellation fee.
BR-015	Deleted records with financial or audit history are archived/soft-deleted; only authorized administrators may purge after retention checks.
BR-016	Cost totals exclude drafts and voided entries, include posted adjustments, and expose tax-inclusive and tax-exclusive views.
BR-017	When included token units reset, unused carryover is applied only if the plan version explicitly allows it.
BR-018	Shared workspace totals must respect row-level ownership/visibility and never leak one user's masked payment data to another.
BR-019	All dates, currencies, and numeric inputs are validated server-side; browser formatting is not authoritative.
BR-020	Alert delivery failure is recorded and retried using controlled backoff; a failed delivery does not mark an alert acknowledged.

4.4 Cost Calculation Rules

Metric	Calculation / interpretation
Normalized monthly recurring cost	$\text{Net recurring amount} \times 12 / \text{billing interval in months}$; annual plans divide by 12.
Normalized annual recurring cost	$\text{Normalized monthly recurring cost} \times 12$.
Net cost entry	$\text{Gross charge} + \text{tax} + \text{surcharge} - \text{discount} - \text{credit} - \text{refund}$.
On-demand actual	Posted metered usage evaluated against the effective pricing tier plus fixed/minimum charges.
Run-rate forecast	$\text{Actual period-to-date} / \text{elapsed billable days} \times \text{total billable days}$, bounded by configured minimum/maximum if present.
Hybrid forecast	Base fee + forecast overage after included units and carryover.
Forecast period total	$\text{Actual posted} + \text{expected committed not yet posted} + \text{forecast variable usage} + \text{expected taxes/fees}$.
Target utilization	$\text{Comparison basis amount} / \text{target amount} \times 100$; undefined when target equals zero.

Metric	Calculation / interpretation
Effective token cost	Allocated net package cost / consumed paid units; free/granted units shown separately.
Savings opportunity	Projected avoidable future cost after cancellation deadline, excluding sunk cost and mandatory fees.
FX conversion	Original amount × stored conversion rate to base currency; inverse rate is clearly identified if used.

ROUNDING Perform calculations using database decimals at provider-supported precision. Round only for display or final currency posting. Totals must equal the sum of stored base-currency entries within a configurable minor-unit tolerance.

5. Future-State Processes

5.1 Subscription Lifecycle

Discover / import → validate vendor and tool → select subscription type → record plan and payment account → set targets and alert policy → activate / trial → record charges and usage → review forecast → renew, change, pause, or cancel → retain history.

5.2 Monthly Cost-Control Cycle

1. At period start, create expected commitments from active plan schedules and carry forward approved targets.
2. Ingest or manually record recurring fees, token purchases, usage, credits, refunds, taxes, and adjustments.
3. Reconcile entries to subscriptions and payment accounts; route unmatched entries to an exception queue.
4. Recalculate actual, committed, and forecast totals by category, tool, payment source, and target scope.
5. Evaluate warning and critical thresholds; create deduplicated alerts and escalation tasks.
6. Review the dashboard, acknowledge alerts, correct missing data, and take renewal/cancellation actions.
7. Close the period, lock the management snapshot, and compare actual versus target and prior periods.

5.3 Renewal and Trial Review

1. Scheduler identifies active trials, renewal/cancellation deadlines, price reviews, and payment expiries in configured windows.
2. System issues reminders with tool, amount, payment account alias, deadline, forecast impact, and direct action link.
3. User chooses renew, change plan, cancel, pause, snooze, or record a decision note.
4. The plan schedule and forecast update from the effective date; historical entries remain unchanged.

5.4 On-Demand and Token Control

1. Define meter unit, pricing tiers, included units, budget target, forecast method, and data-refresh expectation.
2. Record usage manually, via file import, or through an approved connector; detect duplicates by provider reference and period.
3. Apply included units, carryover, package balances, tier rates, minimums, and caps in effective-date order.
4. Compare consumption and cost against target; alert on rapid growth, stale data, depleted balance, or forecast overrun.

6. Functional Requirements

Priority uses MoSCoW: Must is required for the initial production release; Should is strongly expected; Could may be scheduled after core acceptance. Each requirement is testable and traceable to at least one use case or acceptance scenario.

6.1 Identity, Workspace and Preferences

ID	Requirement	Priority	Acceptance summary
FR-001	Authenticate with email/password and optional approved social/SSO provider.	Must	Valid users sign in; invalid and locked users receive safe errors.
FR-002	Support email verification, password reset, session logout, and configurable inactivity timeout.	Must	Security flows complete with expiring, single-use tokens.
FR-003	Maintain profile, base currency, locale, time zone, date/number format, fiscal month, and week start.	Must	All views and scheduled actions honor saved preferences.
FR-004	Support owner, administrator, editor, viewer/auditor, and optional finance-reviewer roles.	Must	Each protected action is permitted or denied server-side by role.
FR-005	Allow dashboard, default period, column, density, and notification preferences.	Should	Preferences persist across devices and can be reset.
FR-006	Provide account data export and controlled deactivation/closure workflow.	Should	User receives export and retention warning before closure.

6.2 Tool, Vendor, Category and Subscription Type Management

ID	Requirement	Priority	Acceptance summary
FR-010	Create, edit, archive, search, filter, tag, and merge duplicate vendors and tools.	Must	Records preserve linked subscription and history references.
FR-011	Store tool name, vendor, category, purpose, URL, logo/icon, notes, owner, environment, tags, and status.	Must	Required fields validate and display consistently.
FR-012	Manage configurable categories and subcategories with color/icon and active state.	Must	Categories can be reused in filters, targets, and reports.
FR-013	Manage subscription types and characteristics: recurring, prepaid, usage, trial, free, one-time, hybrid.	Must	Type rules drive relevant form fields and calculations.
FR-014	Prevent deletion of a type/category in use; allow replacement mapping and archive.	Must	System offers remap or blocks with impact count.
FR-015	Support custom fields for tool/subscription metadata with type, validation, visibility, and order.	Could	Configured fields render and validate without code change.

6.3 Subscription Lifecycle and Plan Versioning

ID	Requirement	Priority	Acceptance summary
FR-020	Create subscriptions manually, by duplication, guided onboarding, or validated import.	Must	A complete active subscription can be saved in under three screens.
FR-021	Capture plan name, status, start/end, trial end, billing cadence, billing anchor, renewal, cancellation deadline, grace period, and auto-renew.	Must	Lifecycle dates produce correct reminders and schedule.

ID	Requirement	Priority	Acceptance summary
FR-022	Support monthly, annual, interval-based, prepaid, on-demand, hybrid, free, trial, and one-time pricing.	Must	Relevant calculations match the selected model.
FR-023	Version plan price, currency, entitlements, included units, tax, discount, and payment account by effective date.	Must	Past costs retain prior version; future forecast uses current version.
FR-024	Support add-ons, seats, environments, and multiple cost components under one subscription.	Should	Components roll into totals and remain individually reportable.
FR-025	Support pause, resume, cancel, expire, reactivate, and archive with reason and effective date.	Must	Status transition rules prevent invalid or backdated overlap.
FR-026	Record vendor account reference, login URL, owner, recovery contact, and attachment without storing passwords.	Should	Sensitive secret fields are not available.
FR-027	Maintain notes, decision history, cancellation evidence, invoice/receipt attachments, and renewal outcome.	Should	Authorized users can retrieve the evidence from history.
FR-028	Show next charge, renewal, cancellation deadline, days remaining, and normalized cost.	Must	Values update after any schedule or plan change.

6.4 Cost Ledger, Recurring Charges and Adjustments

ID	Requirement	Priority	Acceptance summary
FR-030	Maintain a ledger of draft, posted, voided, refunded, and reconciled cost entries.	Must	Only posted/eligible entries affect totals per business rules.
FR-031	Record charge type, service period, transaction date, due date, amount, tax, discount, credit, currency, FX rate/source, vendor reference, and notes.	Must	Entry preserves original and base-currency values.
FR-032	Generate expected recurring commitments from subscription schedules without duplicating posted charges.	Must	Matched actuals replace/offset expected commitments once.
FR-033	Support proration, partial periods, minimum charges, setup fees, late fees, refunds, credits, and manual adjustments.	Must	Net totals and audit references calculate correctly.
FR-034	Import cost entries from CSV/XLSX with mapping, preview, validation, duplicate detection, and error export.	Should	Valid rows import atomically or by clear partial-result policy.
FR-035	Reconcile an entry to tool, subscription, cost component, payment account, and statement reference.	Should	Matched and unmatched totals are visible.
FR-036	Permit authorized correction by reversal and replacement, not silent historical overwrite.	Must	Original, reversal, replacement, actor, and reason are linked.
FR-037	Close and reopen reporting periods under permission with a reason.	Should	Closed periods block ordinary edits and preserve snapshot totals.

6.5 AI Token, Credit and On-Demand Usage

ID	Requirement	Priority	Acceptance summary
FR-040	Define meter units such as input tokens, output tokens, requests, images, minutes, credits, or custom units.	Must	Units retain precision and are labeled in all views.
FR-041	Define flat, tiered, volume, package, included-unit, minimum, capped, and hybrid rates by effective date.	Must	Pricing engine applies the correct rate version and tier logic.
FR-042	Record usage by model/service, environment/project, date range, unit type, quantity, and provider reference.	Must	Duplicates are detected and totals can be segmented.

ID	Requirement	Priority	Acceptance summary
FR-043	Track token/credit packages, granted units, consumed units, remaining balance, carryover, and expiry.	Must	Balance ledger reconciles purchases, grants, use, and expiry.
FR-044	Allocate package consumption using configurable FIFO-by-expiry default and manual override with reason.	Should	Allocation and effective cost are reproducible.
FR-045	Calculate actual usage cost, effective unit cost, included-unit utilization, and estimated overage.	Must	Results match rate and balance ledgers within tolerance.
FR-046	Select on-demand forecast method: target, run-rate, recent average, manual, or zero.	Must	Method and assumptions display beside forecast.
FR-047	Alert on rapid usage growth, forecast overrun, low balance, imminent expiry, and stale usage data.	Must	Each event follows policy, deduplication, and acknowledgement.
FR-048	Support vendor API connectors through encrypted credentials/reference when separately approved.	Could	Connector failures do not block manual entry and are audited.

6.6 Payment Account Management

ID	Requirement	Priority	Acceptance summary
FR-050	Create card, bank, wallet, virtual card, vendor balance, invoice/manual, reimbursement, and custom payment account types.	Must	Required fields adapt to type.
FR-051	Store friendly name, provider/issuer, owner, masked identifier/last four, billing currency, expiry month/year, status, limits, notes, and optional icon.	Must	No full PAN/CVV/password is accepted or displayed.
FR-052	Link a payment account to subscriptions, cost entries, and optional default category/vendor rules.	Must	Impact list is shown before status change.
FR-053	Alert before account expiry, limit threshold, or inactive-account renewal conflict.	Must	Alert includes affected subscriptions and action link.
FR-054	Replace or deactivate an account with guided bulk reassignment and effective date.	Must	Future charges move; historical charges retain original account.
FR-055	Show spend, forecast, subscriptions, next charges, currency exposure, and exceptions per payment account.	Should	Totals reconcile with ledger filters.

6.7 Targets, Budgets, Forecasts and Alerts

ID	Requirement	Priority	Acceptance summary
FR-060	Create global, category, tool, subscription, token/on-demand, and payment-account targets.	Must	Scope is unambiguous and overlapping targets are allowed with labels.
FR-061	Support monthly, quarterly, annual, rolling, custom-date, and one-time target periods.	Must	Period boundaries use workspace time zone and fiscal settings.
FR-062	Set amount, currency, comparison basis, warning/critical percentages or absolute values, and carry-forward rule.	Must	Target utilization is deterministic and displayed.
FR-063	Forecast committed and variable cost using actuals, schedules, usage method, trial conversion, and known price changes.	Must	Forecast explains components and last-calculated time.
FR-064	Create policies for target, renewal, trial, cancellation deadline, payment expiry, stale data, balance, and unusual usage.	Must	Policies can be enabled, tested, and scoped.
FR-065	Deliver in-app and email alerts with optional future channels, quiet hours, digest, escalation, and recipients.	Must	Delivery status and retry history are visible.

ID	Requirement	Priority	Acceptance summary
FR-066	Deduplicate alerts, escalate at higher severity, and allow acknowledge, snooze, dismiss, assign, and resolve.	Must	State changes are audited and do not hide unresolved risk.
FR-067	Provide notification test and preview using sample data.	Should	No real alert is sent unless user confirms test delivery.
FR-068	Recommend a target from historical actuals plus user-selected tolerance without changing it automatically.	Could	Recommendation shows method, period, and confidence limitations.

6.8 Dashboard, Search, Calendar and Reporting

ID	Requirement	Priority	Acceptance summary
FR-070	Show dashboard KPIs for actual, committed, forecast, target variance, active tools, upcoming renewals, trials, alerts, and payment risks.	Must	Cards use the same filter/period context and drill down.
FR-071	Show cost trend, category mix, subscription-type mix, top tools, payment-source mix, token trend, and savings opportunities.	Must	Charts include accessible table/text alternatives.
FR-072	Provide global search and list filters for status, type, category, vendor, owner, payment account, date, amount, tag, alert, and data freshness.	Must	Filters combine, persist in URL/state, and can be cleared.
FR-073	Provide list, compact card, calendar, and timeline views for subscriptions and renewals.	Should	View choice persists and remains usable on mobile.
FR-074	Drill from totals to underlying subscriptions, entries, usage, and calculation explanation.	Must	Every displayed total is explainable.
FR-075	Export filtered data and standard reports to CSV/XLSX and printable/PDF-ready view.	Should	Export matches filter context and masks protected fields.
FR-076	Schedule periodic email digests and reports.	Could	Schedule can be paused and delivery logged.
FR-077	Allow saved views with name, filters, columns, sort, density, visibility, and default flag.	Should	Users can restore and delete their saved views.

6.9 Administration, Audit, Import and Operations

ID	Requirement	Priority	Acceptance summary
FR-080	Manage users, roles, workspace settings, reference data, feature flags, templates, and supported currencies.	Must	Admin actions are permission-checked and audited.
FR-081	Maintain immutable audit events for sign-in, CRUD, status, plan, cost, target, account, alert, import, export, and permission changes.	Must	Audit includes actor, time, action, entity, before/after summary, and context.
FR-082	Provide audit search/export with date, actor, entity, action, outcome, and correlation ID filters.	Should	Results respect retention and access rules.
FR-083	Provide onboarding wizard and sample-data option with safe removal.	Should	A new user can reach a useful dashboard quickly.
FR-084	Provide import templates, mapping profiles, validation preview, result summary, and rollback for failed batch.	Should	Import results are reproducible and errors actionable.
FR-085	Expose background job status for schedule generation, forecast, alerts, email, import, export, and connectors.	Must	Failures surface to administrators with retry controls.
FR-086	Support logical backup/restore procedures and environment health checks.	Must	Recovery is tested and documented.

ID	Requirement	Priority	Acceptance summary
FR-087	Provide configurable data retention and purge workflow with legal/financial hold exceptions.	Should	Purge preview identifies affected records and dependencies.

7. Detailed Use Cases

Use cases describe user intent and externally observable behaviour. Screen names are indicative; the final information architecture may consolidate steps while preserving all validations and outcomes.

UC-01 — Complete first-time setup

Field	Detail
Primary actor	Account Owner
Goal	Create a usable workspace and cost-control baseline.
Preconditions	User has verified an account and has not completed onboarding.
Trigger	User signs in for the first time.
Postconditions	Preferences and valid records are saved; onboarding state is recorded.
Related requirements	FR-003, FR-020, FR-051, FR-060, FR-083, FR-084

Main success flow

1. System asks for base currency, time zone, locale, fiscal-month start, and notification preferences.
2. User optionally loads safe sample data or chooses a blank workspace.
3. User adds at least one payment account alias without entering sensitive card credentials.
4. User adds or imports the first subscription and selects its type.
5. System previews normalized cost, next renewal, and recommended starter target.
6. User confirms settings and reaches the dashboard with an onboarding checklist.

Alternate / exception flows

- User skips optional steps; system retains checklist and uses disclosed defaults.
- Import contains invalid rows; system shows row-level errors without creating partial ambiguous records.

UC-02 — Register a monthly or annual subscription

Field	Detail
Primary actor	Owner / Editor
Goal	Track a recurring tool and its future obligations.
Preconditions	Tool and subscription type exist; user has edit permission.
Trigger	User selects Add subscription.
Postconditions	Active subscription is visible in lists, forecast, calendar, payment account, and reports.
Related requirements	FR-010–FR-014, FR-020–FR-023, FR-028, FR-032, FR-053

Main success flow

1. User selects or creates vendor and tool.
2. User selects monthly, annual, or custom recurring type.
3. User enters plan, amount, currency, billing anchor, renewal, auto-renew, cancellation deadline, and tax/discount.
4. User selects payment account or records a permitted payment exception.
5. System shows cash schedule, normalized monthly/annual cost, and reminder dates.
6. User confirms; system creates subscription, plan version, commitments, audit entry, and alert events.

Alternate / exception flows

- Billing date is invalid for a month; system applies the disclosed end-of-month rule.
- A likely duplicate exists; system warns and offers view/continue.

UC-03 — Register an on-demand AI service

Field	Detail
Primary actor	Owner / Editor
Goal	Track variable usage and prevent budget overrun.
Preconditions	Tool exists and on-demand or hybrid type is active.
Trigger	User selects an on-demand subscription model.
Postconditions	Usage can be posted and cost/forecast evaluated using the effective pricing configuration.
Related requirements	FR-040–FR-047, FR-060–FR-067

Main success flow

1. User selects meter units and defines flat/tiered/package or hybrid rates.
2. User enters included units, minimums, caps, billing period, and data source.
3. User selects forecast method and target amount.
4. User configures warning, critical, rapid-growth, low-balance, and stale-data alerts.

5. System validates rate continuity, displays examples, and saves the configuration.

Alternate / exception flows

- Rate tiers have gaps/overlap; system blocks activation and highlights the affected range.
- No target is entered; system allows activation only after the user explicitly accepts no budget alert.

UC-04 — Record token package purchase and consumption

Field	Detail
Primary actor	Owner / Editor
Goal	Maintain an auditable balance and effective unit cost.
Preconditions	Prepaid token/credit subscription exists.
Trigger	User records a package purchase or imports usage.
Postconditions	Balance ledger and cost ledger reconcile; alerts are evaluated.
Related requirements	FR-031, FR-040–FR-047, BR-007, BR-017

Main success flow

1. User enters package units, paid/free split, net cost, currency, purchase date, and expiry.
2. System posts the purchase and creates balance lots.
3. User enters or imports consumption by service/model and period.
4. System detects duplicates, applies included units and FIFO-by-expiry allocation, and calculates cost.
5. User reviews allocations, remaining balance, effective unit cost, and expiry risk.

Alternate / exception flows

- Consumption exceeds available balance; system records eligible overage or holds the excess as an exception.
- User overrides allocation; system requires a reason and writes audit history.

UC-05 — Change plan or price

Field	Detail
Primary actor	Owner / Editor
Goal	Apply a future or immediate plan change without corrupting history.
Preconditions	Subscription is active/trial/paused and user can edit.
Trigger	Vendor changes price, cadence, currency, entitlements, or payment account.
Postconditions	Historical records use old version; current/future calculations use the new version.
Related requirements	FR-023, FR-028, FR-032, FR-036, FR-063, FR-081

Main success flow

1. User opens Change plan and enters effective date and revised terms.

2. System validates date overlap and previews affected commitments, targets, and forecast.
3. User confirms; system closes the previous version and opens the new version.
4. Future expected commitments and alerts are regenerated.
5. System records before/after summary and reason in the audit trail.

Alternate / exception flows

- Change is backdated into a closed period; system blocks or routes to authorized correction workflow.
- A posted charge differs from expected price; system preserves it and flags reconciliation variance.

UC-06 — Record and reconcile a charge

Field	Detail
Primary actor	Owner / Finance Reviewer
Goal	Post an actual cost and match it to the correct obligation.
Preconditions	Subscription or unmatched-cost queue exists.
Trigger	User enters/imports a charge.
Postconditions	Ledger, commitment, payment account, forecast, target, and audit views are updated once.
Related requirements	FR-030–FR-036, FR-060–FR-066

Main success flow

1. User provides transaction/service dates, amount, tax, discount, currency, payment account, and reference.
2. System proposes subscription and expected commitment matches.
3. User accepts or changes match and reviews FX conversion.
4. System posts the entry, offsets the matched commitment, and recalculates totals.
5. If target severity changes, the system creates or escalates an alert.

Alternate / exception flows

- Duplicate reference/amount/date detected; system blocks or requires explicit override.
- No match exists; entry remains posted/unmatched and appears in reconciliation exceptions.

UC-07 — Create a target and alert policy

Field	Detail
Primary actor	Owner / Editor
Goal	Control spending for a defined scope and period.
Preconditions	At least one eligible scope exists.
Trigger	User selects Set target.
Postconditions	Target appears on dashboards and evaluates whenever relevant cost data changes.
Related requirements	FR-060–FR-067

Main success flow

1. User selects global/category/tool/subscription/token/payment-account scope.
2. User selects period, amount, currency, comparison basis, and carry-forward rule.
3. User enters warning and critical thresholds and chooses channels/recipients.
4. System previews current utilization and example alert timing.
5. User activates the target and policy.

Alternate / exception flows

- Thresholds are reversed or outside allowed range; system blocks save.
- An overlapping target exists; system explains overlap and allows intentional coexistence.

UC-08 — Respond to a target-crossing alert

Field	Detail
Primary actor	Owner / Assignee
Goal	Understand and resolve a spend risk.
Preconditions	Active target crossed a configured threshold.
Trigger	Alert is delivered in-app or by email.
Postconditions	Alert state, action history, and target status are updated without duplicate notifications.
Related requirements	FR-063–FR-066, FR-074, FR-081

Main success flow

1. User opens the alert and sees severity, amount, variance, forecast, scope, and contributing costs.
2. User drills into underlying subscriptions/usage and calculation explanation.
3. User chooses acknowledge, assign, snooze, edit target, correct data, reduce usage, or cancel/change a plan.
4. System records the action and reevaluates severity after any data change.
5. User resolves the alert with a note when the risk is addressed or formally accepted.

Alternate / exception flows

- Email delivery fails; in-app alert remains active and retry status is visible.
- Forecast drops below threshold; system records automatic recovery but retains the event history.

UC-09 — Review renewal and cancel a subscription

Field	Detail
Primary actor	Owner / Editor
Goal	Avoid an unwanted renewal and retain evidence.
Preconditions	Active subscription has renewal/cancellation dates.
Trigger	Renewal reminder is received or calendar item opened.
Postconditions	Subscription lifecycle and forecast reflect the confirmed commercial outcome.
Related requirements	FR-021, FR-025, FR-027, FR-028, FR-033, FR-064–FR-066

Main success flow

1. User reviews price, utilization, recent cost, forecast, payment account, and cancellation deadline.
2. User records decision to cancel, effective/access-end date, reason, and optional evidence attachment.
3. System previews avoided future cost, outstanding fees, and affected alerts.
4. User confirms cancellation.
5. System updates status/schedule, stops eligible future commitments, retains history, and records outcome.

Alternate / exception flows

- Cancellation is after deadline; system warns that renewal may still occur and keeps the commitment until confirmed.
- Vendor issues refund/fee; user posts linked ledger entry.

UC-10 — Replace an expiring payment account

Field	Detail
Primary actor	Owner / Administrator
Goal	Move future charges safely while retaining payment history.
Preconditions	Existing account is expiring/inactive; replacement account exists.
Trigger	Expiry alert or account action is opened.
Postconditions	Future charges reference the new account; historical charges retain the old masked account.
Related requirements	FR-050–FR-055, FR-081

Main success flow

1. System lists affected subscriptions, next charges, currencies, and renewal dates.
2. User selects all or a subset and chooses replacement effective date.
3. System validates replacement status/currency and previews impact.
4. User confirms; future plan versions reference the replacement account.

5. Original account is marked inactive when no future obligations remain.

Alternate / exception flows

- Some subscriptions must be updated at vendor portal; system creates follow-up actions rather than claiming completion.
- User lacks access to a linked subscription; bulk update excludes it and reports the exception.

UC-11 — Import subscriptions, usage or costs

Field	Detail
Primary actor	Owner / Editor
Goal	Create or update records efficiently from a file.
Preconditions	User has a supported file and import permission.
Trigger	User opens Import.
Postconditions	Import batch, row outcomes, audit events, and recalculated totals are available.
Related requirements	FR-020, FR-034, FR-042, FR-084, FR-085

Main success flow

1. User selects entity type, downloads template or uploads CSV/XLSX, and chooses a saved mapping.
2. System scans structure, shows mapping, normalizes values, and previews valid/warning/error rows.
3. User resolves required mappings and duplicate decisions.
4. System executes the approved batch and reports created, updated, skipped, and failed rows.
5. User downloads error file and can rerun corrected rows using the batch reference.

Alternate / exception flows

- Fatal validation or malware scan failure prevents processing.
- Background job fails; system leaves batch retryable and avoids duplicate committed rows.

UC-12 — Analyze dashboard and drill into cost

Field	Detail
Primary actor	Owner / Viewer
Goal	Understand current cost position and the reason for each total.
Preconditions	User has view permission and records exist.
Trigger	User opens Dashboard or a saved view.
Postconditions	User can explain the total and take a relevant follow-up action.
Related requirements	FR-070–FR-077

Main success flow

1. System loads actual, committed, forecast, targets, alerts, renewals, token usage, and payment risks for the selected period.
2. User changes period, currency view, owner/category/status filters, or saved view.
3. Charts and KPIs update consistently and preserve filter context.
4. User selects a KPI/chart segment to open the contributing records and calculation explanation.
5. User exports or shares a masked report if authorized.

Alternate / exception flows

- No data exists; system shows a helpful empty state and onboarding actions.
- Some source data is stale; system marks freshness and excludes/estimates according to disclosed rules.

UC-13 — Correct a posted historical cost

Field	Detail
Primary actor	Finance Reviewer / Administrator
Goal	Correct an error while preserving financial and audit integrity.
Preconditions	Posted entry exists and user has correction permission.
Trigger	User selects Correct entry.
Postconditions	No original evidence is lost; reports use the valid posted chain.
Related requirements	FR-036, FR-037, FR-081

Main success flow

1. System displays original entry and dependent matches, totals, and targets.
2. User enters correction reason and replacement values.
3. System creates a linked reversing entry and a new corrected entry.
4. System recalculates affected periods, snapshots, targets, and alerts according to close rules.
5. Audit view shows the entire correction chain.

Alternate / exception flows

- Period is closed; authorized user must reopen with reason or post an adjustment in the current period.
- Correction would unmatched a refund/credit; system requires resolving the dependency first.

UC-14 — Manage subscription types and reference data

Field	Detail
Primary actor	Administrator
Goal	Keep configuration relevant without breaking existing data.
Preconditions	Administrator is authenticated.
Trigger	Admin opens Settings / Reference data.
Postconditions	Reference configuration is active, traceable, and backward compatible.
Related requirements	FR-012–FR-015, FR-080–FR-082

Main success flow

1. Administrator creates or edits a subscription type/category with display order, active state, and applicable cost characteristics.
2. System validates unique keys and configuration consistency.
3. Administrator previews affected records before archive or merge.
4. System blocks destructive deletion and offers replacement mapping.
5. System logs the change and applies it to new/eligible forms.

Alternate / exception flows

- A type is locked because it is system-required; admin may rename display label but not alter core semantics.
- Mapping would create invalid data; system reports affected records and stops.

UC-15 — Export data and close the account

Field	Detail
Primary actor	Account Owner
Goal	Obtain a portable copy and deactivate service safely.
Preconditions	Owner is authenticated and passes step-up verification if configured.
Trigger	User opens Privacy / Account.
Postconditions	Account is deactivated, data disposition is scheduled, and evidence is retained as authorized.
Related requirements	FR-006, FR-075, FR-081, FR-085, FR-087

Main success flow

1. User requests a data export and selects included scopes.
2. System prepares a masked, time-limited export and records the event.
3. User reviews active renewals, retention, legal hold, and consequences.
4. User confirms deactivation/closure with reason and verification.

5. System revokes active sessions, stops notifications/jobs as appropriate, and applies retention workflow.

Alternate / exception flows

- Export job fails; user is notified and may retry without duplicate files.
- Retention hold prevents purge; system explains the retained categories and deactivation continues.

8. User Experience and Responsive UI Requirements

8.1 UX Principles

- Compact, not cramped: prioritize current cost, next risk, and next action; reveal detail progressively.
- One calculation language: Actual, Committed, Forecast, Target, and Variance use consistent labels and colors everywhere.
- Explain every number: totals link to contributing records, formula basis, data freshness, currency conversion, and exclusions.
- Fast entry: sensible defaults, conditional fields, duplicate/quick-add, keyboard support, inline validation, and saved mappings.
- Safe actions: preview financial impact, confirm destructive lifecycle actions, support undo where safe, and retain history.
- Responsive by design: the same capability is available on mobile, but layout adapts to touch, narrow width, and short sessions.
- Accessible and calm: color is never the only signal; alerts are prioritized; empty, loading, stale, error, and offline states are explicit.

8.2 Information Architecture

Primary navigation	Key destinations
Dashboard	Overview, targets, alerts, spend trend, renewals, usage, payment risks, quick actions.
Subscriptions	All subscriptions, tools, vendors, calendar, trials, renewals, archived records.
Usage & Tokens	Meters, usage entries, packages/balances, imports, models/projects, forecasts.
Costs	Ledger, expected commitments, reconciliation, adjustments, closed periods, imports.
Targets & Alerts	Targets, policies, alert inbox, notification history, recommendations.
Payment Accounts	Cards/accounts/wallets, linked subscriptions, expiry, limits, reassignment.
Reports	Cost summary, variance, renewal, utilization, payment source, token, savings, exports.
Settings	Profile, workspace, categories, subscription types, roles, imports, integrations, audit, operations.

8.3 Screen Catalogue and UI Acceptance

Screen	Essential content	Primary actions	Responsive behaviour
Sign in / Recovery	Email, password/SSO, verification and safe errors	Sign in, reset, verify	Single column; large touch targets; password manager compatible.
Onboarding	Progress, preferences, payment alias, first subscription, target	Back, skip, save, finish	Step layout becomes stacked; sticky bottom action bar.
Dashboard	KPI strip, target bars, alerts, renewals, charts, quick actions	Filter, drill, add, export	Cards stack 1-up; charts have compact/table view; filters in sheet.
Subscription List	Name, type, status, owner, cost, renewal, payment alias, alert	Search, filter, sort, bulk action	Desktop table switches to compact cards or priority columns.
Subscription Detail	Overview, plan history, costs, usage, targets, alerts, attachments, audit	Edit, change plan, pause, cancel	Tabs become horizontally scrollable or select menu; actions in bottom sheet.

Screen	Essential content	Primary actions	Responsive behaviour
Add/Edit Subscription	Conditional form by type with live cost preview	Save draft, validate, activate	One-column sections; numeric keypad; dates and selects touch friendly.
Usage & Token Detail	Balances, expiry, rate tiers, consumption, cost, forecast	Add/import usage, buy package, adjust	Summary fixed above collapsible detailed tables.
Cost Ledger	Posted/draft/expected entries, matching and variance	Add, import, reconcile, correct	Priority columns plus row detail drawer; bulk tools in overflow.
Target Editor	Scope, period, amount, basis, thresholds, channels, preview	Test, activate, duplicate	Form stacks; preview stays visible as compact summary.
Alert Inbox / Detail	Severity, state, variance, contributors, delivery, actions	Acknowledge, assign, snooze, resolve	List cards; action buttons remain reachable by thumb.
Payment Account Detail	Masked identity, expiry, spend, linked plans, next charges	Edit, replace, deactivate	No sensitive data exposed; linked items rendered as cards.
Reports	Report selector, filters, summary, chart/table, freshness	Run, save, export, schedule	Filters in sheet; printable rendering remains desktop-width.
Settings / Admin	Reference data, roles, alerts, audit, operations	Create, edit, archive, test	Grouped navigation; avoid very wide editable grids.

8.4 Dashboard Composition

- Top bar: workspace, global search, period selector, currency view, notifications, profile, and Add button.
- KPI row: Actual MTD, Committed Remaining, Forecast EOM, Target Variance, Next 30-Day Renewals, and Active Critical Alerts.
- Priority area: target progress, critical/warning alert list, renewal/trial timeline, and payment-expiry conflicts.
- Insight area: cost trend, category/tool mix, recurring versus on-demand, token cost/usage, and top increases.
- Action area: add subscription, record cost, record usage, buy package, create target, import, and run reconciliation.
- Every tile shows period, currency, data freshness, accessible label, and drill-down destination; ambiguous standalone percentages are prohibited.

8.5 Responsive Breakpoints and Layout

Viewport	Layout requirements
≥1200 px	Persistent sidebar; 12-column grid; dense table view; multi-panel detail where useful.
992–1199 px	Collapsible sidebar; 8/12-column grid; reduced secondary columns; no horizontal page scroll.
768–991 px	Drawer navigation; 2-column cards; tables use priority columns and row details.
≤767 px	Single-column content; bottom/sticky primary actions; filters in full-height sheet; cards replace wide tables.
≤375 px	Core add/edit/alert tasks remain operable; labels wrap; touch targets do not overlap; charts offer text/table alternative.

8.6 Form, Table and State Requirements

- Required fields are identified in label text; server and client validation use the same rule and specific corrective messages.
- Conditional fields appear only for the selected subscription/cost model, while changing type warns before discarding entered values.
- Money inputs show currency, accept appropriate decimal precision, preserve unrounded value, and never use floating-point arithmetic server-side.
- Dates show time-zone context when timing matters; renewal, cancellation deadline, and access-end date remain distinct.
- Tables have sticky headers on desktop, column chooser, sort, filter, pagination or virtualization, accessible row actions, and export matching filters.
- All data views define loading skeleton, no-data, no-results, partial/stale data, permission denied, recoverable error, and fatal error states.
- Bulk actions show selected count and an impact preview; sensitive or destructive bulk changes require explicit confirmation.
- Success notices name the saved object and provide a useful next action; errors preserve entered data.

8.7 Accessibility and Usability

- Meet WCAG 2.2 AA as the product target, subject to formal accessibility validation before release.
- All functions operate by keyboard with visible focus, logical order, skip links, and no keyboard traps.
- Touch targets are at least 44×44 CSS pixels where practical; compact density never reduces critical action targets below accessible size.
- Text and meaningful UI components meet contrast requirements; warning/critical states use icon, label, and text in addition to color.
- Charts have descriptive titles, accessible summaries, tooltips reachable by keyboard, and equivalent data tables.
- Modals, drawers, menus, tabs, toasts, and validation messages expose correct accessible names, roles, state, and focus restoration.
- The interface supports 200% text zoom and browser zoom without loss of core functionality or two-dimensional scrolling, except genuine data tables.

8.8 Metronic Application Guidance

Use Metronic as the visual/component foundation, not as the source of business rules. Establish an application design layer that maps Metronic tokens and components to approved semantic roles: primary action, neutral action, success, warning, critical, disabled, and focus. Remove unused demo assets, pages, plugins, and example credentials. Wrap vendor components so accessibility, validation, authorization, responsive behaviour, and future template upgrades remain controllable by the application.

9. Data Requirements and Logical Data Model

9.1 Logical Entity Catalogue

Entity	Purpose / key data	Primary relationship
Workspace	Base currency, locale, time zone, fiscal settings, feature and retention policy.	1:N users, tools, targets
User / Role	Identity, status, preferences, role memberships.	N:M workspace roles
Vendor	Supplier identity, URL, support and billing metadata.	1:N tools
Tool	Product identity, category, purpose, owner, tags, status.	1:N subscriptions
Category / Tag	Configurable classification and presentation metadata.	N:M tools/subscriptions
Subscription Type	Commercial model and enabled characteristics.	1:N subscriptions
Subscription	Lifecycle, owner, plan, status, dates, vendor account reference.	1:N plan versions
Plan Version	Effective terms, billing cadence, amount, tax/discount, auto-renew, payment account.	1:N cost components/rates
Cost Component	Base fee, seat, add-on, usage, tax, surcharge, discount semantics.	1:N schedules/entries
Rate / Tier	Meter unit, range, price, minimum/cap, included units, effective date.	N:1 plan version
Usage Entry	Period, project/model, units, source, provider reference, status.	N:1 meter/subscription
Token Package / Lot	Purchased/granted units, cost, currency, expiry and balance.	1:N allocations
Usage Allocation	Units allocated from a package/included/overage source.	N:1 usage and lot
Cost Entry	Ledger event, original/base amounts, dates, status, references.	N:1 subscription/account
Expected Commitment	Scheduled future obligation and matching state.	N:1 plan component
Payment Account	Masked payment source, owner, currency, expiry, status.	1:N plans/entries
Target	Scope, period, amount, basis, thresholds, status.	1:N evaluations
Alert Policy	Event type, scope, thresholds, channels, recipients and schedule.	1:N alerts
Alert / Delivery	Event state, severity, message, deduplication, acknowledgement and channel outcome.	N:1 policy
Attachment	Safe file metadata, storage key, hash, classification.	Polymorphic authorized link
Import Batch / Row	Mapping, file, counts, validation and processing outcome.	1:N staged rows
Audit Event	Actor, action, entity, correlation, outcome, before/after summary.	Append-only

9.2 Data Integrity Requirements

- Use surrogate primary keys and workspace/tenant keys on all owned records; use unique constraints for business identifiers within scope.
- Use DECIMAL for money, FX, rates, and high-precision usage; never use FLOAT/DOUBLE for financial values.
- Store date-only commercial deadlines separately from UTC timestamps; store workspace time zone used for interpretation.
- Plan/rate effective ranges must not overlap for the same commercial component unless explicitly modeled as additive.

- Ledger totals derive from immutable posted entries and linked reversals; cached summaries are rebuildable from source transactions.
- Foreign keys, state constraints, idempotency keys, provider references, and unique batch-row keys prevent orphaning and duplication.
- Every imported or automated record retains source, batch/connector, provider reference, received time, and original payload hash/trace as permitted.
- Soft deletion is used for records with historical references; purge is a separate permissioned retention operation.

9.3 Data Classification

Class	Examples	Handling
Public/Reference	Tool/vendor public name, category labels	Normal access within product context.
Internal	Subscription plan, cost, target, notes, usage	Authenticated, role-controlled, encrypted in transit/at rest.
Confidential	Payment aliases, invoice attachments, exports, audit history	Least privilege, masking, access logging, retention controls.
Restricted / prohibited	Full PAN, CVV, bank password, vendor password, raw secret/API token in ordinary fields	Do not collect. Approved connector secrets require dedicated encrypted secret storage and rotation.

9.4 Retention and Archival Baseline

Retention periods are configurable by record class and jurisdiction. As a baseline, keep financial and audit history for the approved management-reporting period, keep import staging files only long enough to validate and recover the batch, and expire generated exports quickly. Purge must verify legal/financial holds, dependencies, backups, and workspace ownership before execution.

10. Solution Architecture and Technical Requirements

10.1 Target Architecture

The solution will be a modular Laravel web application with server-rendered and/or approved reactive UI components using Metronic, a MySQL transactional database, background queues, scheduled jobs, object storage for attachments/exports, cache/lock service, and integrations behind explicit adapters. A modular monolith is preferred for the initial release to preserve transactional consistency and operational simplicity while maintaining clear bounded modules.

Layer / module	Responsibilities
Presentation	Metronic layout and components, responsive views, form requests, accessible charts/tables, API/web endpoints.
Identity & Workspace	Authentication, authorization, preferences, tenant/workspace boundary, session/security events.
Catalogue & Subscription	Vendors, tools, types, plan versions, lifecycle, renewal schedules, attachments.
Cost & Reconciliation	Ledger, expected commitments, imports, matching, FX, close/reopen and corrections.
Usage & Pricing	Meters, rate versions, packages, allocations, usage costing and on-demand forecasting.
Budget & Alerting	Targets, evaluations, forecasts, alert policy, deduplication, escalation and delivery.

Layer / module	Responsibilities
Reporting	Read models, filters, KPI calculations, exports, scheduled reports and data freshness.
Operations & Audit	Jobs, health, integration runs, audit, retention, backups and administration.

10.2 Laravel Implementation Requirements

- Use Form Request or equivalent server-side validation, policies/gates for authorization, service/domain classes for calculations, and database transactions for multi-record financial operations.
- Use migrations, seeders for system reference data, factories for tests, and explicit indexes for workspace, status, dates, references, and report filters.
- Queue email, imports, exports, connectors, forecast recomputation, and heavy reports; use idempotent jobs with retry/backoff and dead-letter visibility.
- Use the scheduler for recurring commitments, renewal/trial/payment checks, target evaluation, data freshness, report delivery, retention, and health checks.
- Treat alerts and audit as domain events/outbox-backed where necessary so database changes and external delivery remain consistent.
- Centralize money, period, recurrence, FX, pricing-tier, and forecast logic in tested domain services; views must not duplicate formulas.
- Implement versioned application interfaces for integrations; use circuit breakers/timeouts and retain manual fallback.
- Use configuration/environment secrets outside source control; production debugging must not expose financial or credential data.

10.3 MySQL Requirements

- Use InnoDB, utf8mb4, explicit foreign keys, UTC timestamps, decimal columns sized for expected amounts/units, and consistent collation.
- Index by workspace plus common filters; verify composite indexes using representative queries and data volumes.
- Use transactions and row/application locks for balance allocation, duplicate-sensitive imports, reconciliation, plan changes, and target state transitions.
- Prefer normalized source-of-truth tables; use summary tables/materialized read models only when refresh and rebuild rules are explicit.
- Partitioning or archival tables may be introduced only after measured volume warrants them and recovery/query behaviour is tested.

10.4 Integration Boundaries

Integration	Direction	Baseline behaviour	Failure behaviour
Email provider	Outbound	Alerts, verification, recovery, digests, reports	Retry; in-app record remains; admin sees failure.

Integration	Direction	Baseline behaviour	Failure behaviour
FX-rate provider	Inbound	Approved dated conversion rates	Use last approved/manual rate with freshness warning; never invent.
Vendor usage/billing API	Inbound	Usage, charges, balances where authorized	Mark stale, retry, and preserve manual/import path.
Object storage / malware scan	Both	Attachments and exports with signed access	Reject unsafe file; no public bucket or permanent URL.
Optional bank/card feed	Inbound	Masked transactions for reconciliation	Connector isolated; no payment initiation; unmatched data retained safely.
Optional notification channel	Outbound	Browser push/SMS/chat after approval	Channel-specific failure logged without suppressing in-app alert.

11. Security, Privacy, Audit and Compliance

11.1 Security Requirements

ID	Requirement
SEC-01	Enforce authentication and server-side authorization for every data access and action.
SEC-02	Separate workspace/tenant data in every query, cache key, job, export, attachment path, and audit search.
SEC-03	Protect against CSRF, XSS, SQL injection, mass assignment, insecure direct-object reference, file upload abuse, and session fixation using framework controls and tests.
SEC-04	Use TLS in transit and platform/database/storage encryption at rest; manage secrets in an approved secret store.
SEC-05	Never store full card number, CVV, banking password, or vendor password; reject these patterns where feasible and educate users.
SEC-06	Mask payment aliases in UI, notifications, logs, exports, screenshots, and support tools according to role.
SEC-07	Rate-limit sign-in, recovery, imports, exports, alert tests, and connector endpoints; detect abuse without leaking account existence.
SEC-08	Use secure cookies, session rotation, expiry, logout-all-devices, and optional MFA/step-up for exports, role changes, or account closure.
SEC-09	Validate MIME/content, size, extension, malware result, ownership, and signed access for attachments and generated files.
SEC-10	Audit sensitive reads/exports and all material writes; protect audit events from ordinary modification.
SEC-11	Log application events with correlation IDs and redaction; logs must exclude secrets, raw sensitive payloads, and unnecessary financial detail.
SEC-12	Patch supported dependencies, scan source/dependencies, review Metronic/plugins, and remediate vulnerabilities under an agreed severity SLA.

11.2 Privacy Requirements

- Collect only data required for subscription management; explain purpose at entry and in privacy notice.
- Support user access/export, correction, deactivation, and deletion/purge where retention obligations permit.
- Separate consent/authorization for vendor, financial-feed, and external notification integrations.

- Define data processor/controller roles, hosting region, subprocessors, breach process, and jurisdiction-specific duties before launch.
- Use synthetic or anonymized data in non-production; production copies require approval and masking.

11.3 Audit Event Minimum Fields

Event ID, workspace, UTC time, actor/service, role, action, outcome, entity type/ID, before/after summary or changed fields, reason where required, IP/device context as permitted, correlation/request ID, import/job/connector reference, and retention class.

12. Non-Functional Requirements

ID	Attribute	Requirement / target
NFR-01	Availability	Target 99.5% monthly for production excluding approved maintenance; final SLA confirmed before launch.
NFR-02	Interactive performance	95th percentile server response under 2 seconds for common list/detail actions at agreed load; dashboard under 3 seconds with cached summaries.
NFR-03	Heavy work	Imports, exports, connector sync, large reports and recalculation run asynchronously with progress and safe retry.
NFR-04	Scale baseline	Support at least 10,000 subscriptions, 1,000,000 ledger/usage rows per workspace, and 100 concurrent active sessions, subject to load-test refinement.
NFR-05	Reliability	Financial operations are atomic/idempotent; duplicate delivery or job retry must not duplicate cost or usage.
NFR-06	Recovery	Baseline RPO ≤24 hours and RTO ≤8 hours; production target refined by sponsor; restore drills at least twice yearly.
NFR-07	Accessibility	WCAG 2.2 AA target for supported core workflows, verified by automated and manual testing.
NFR-08	Responsive compatibility	Current major Chrome, Edge, Firefox, Safari and supported mobile Safari/Chrome; graceful behaviour from 320 px width.
NFR-09	Maintainability	Modular code, automated migrations, static analysis, tests, coding standard, documented calculations and public interfaces.
NFR-10	Observability	Health, structured logs, metrics, queue depth, failed jobs, alert delivery, import/connector outcomes and calculation latency.
NFR-11	Localization	Locale-aware date/number/currency formatting; user text supports Unicode; business calculations remain locale independent.
NFR-12	Data portability	Human-readable and machine-readable exports preserve IDs, dates, currency and relationship references where appropriate.
NFR-13	Supportability	User-facing correlation code, admin diagnostics, safe impersonation only if approved and audited, and runbooks for common failures.
NFR-14	Deployment	Separate environments, automated CI/CD gates, backward-compatible migrations, rollback/roll-forward plan, and seeded reference data.
NFR-15	Time accuracy	NTP-synchronized infrastructure; UTC storage plus explicit workspace time-zone conversion and DST tests.

13. Reporting, Analytics and Notification Catalogue

13.1 Standard Reports

Report	Core measures and dimensions
Executive Cost Summary	Actual, committed, forecast, target, variance, recurring/on-demand, trend; by period, owner, category, vendor, tool.
Subscription Inventory	Status, type, plan, normalized cost, owner, payment account, renewal, cancellation deadline, data freshness.
Renewal & Trial Calendar	Renewal/trial/cancellation/payment dates, amount, risk, decision and owner.
Target Variance	Target scope/period/basis, utilization, warning/critical crossing, contributing costs and actions.
Token & Usage	Units, included/paid/free, package balances, expiry, effective unit cost, forecast and overage.
Payment Account Exposure	Spend, commitments, linked subscriptions, next charges, currency, expiry and exceptions.
Reconciliation	Expected vs posted, matched/unmatched, variance, duplicate candidates, stale imports and closed-period status.
Savings Opportunities	Unused/low-use, duplicates, price increase, cancellation window, projected avoidable cost and confidence.
Audit & Change	Actor, action, outcome, entity, changed fields, reason, correlation and export.

13.2 Common Report Filters

Workspace, reporting period, original/base currency, owner, vendor, tool, category/subcategory, subscription type, status, billing cadence, payment account, cost entry type/status, target scope/severity, tag, renewal window, amount range, data source, reconciliation state, and freshness.

13.3 Notification Catalogue

Event	Default timing / trigger	Minimum message content
Target warning	Warning basis reached	Scope, period, current/forecast, target, variance, contributors, action link.
Target critical	Critical basis reached or warning escalates	Severity, current/forecast, target, gap, owner/assignee, action link.
Renewal due	Configurable 30/14/7/1 days	Tool, plan, amount/currency, renewal, cancellation deadline, payment alias.
Trial ending	Configurable 7/3/1 days	Trial end, expected conversion price, auto-convert status, action choices.
Cancellation deadline	Before contractual deadline	Deadline, renewal amount, decision status and direct review link.
Payment account expiry	90/60/30 days or conflict detected	Masked account, expiry, affected subscriptions and next charges.
Token/credit low	Balance or days-of-use threshold	Remaining units, burn rate, expiry, forecast, package/action link.
Unusual usage	Rate-of-change/anomaly rule	Period, baseline, current usage/cost, source freshness and contributors.
Stale data / connector failure	Expected refresh missed or repeated failure	Source, last success, missing period, retry/manual action.
Import/export/report complete	Job terminal state	Outcome counts, safe file link/expiry or error summary.

13.4 Alert Severity and State

Dimension	Values	Rules
Severity	Info, Warning, Critical	Severity escalates when risk increases; recovery does not erase prior event.
Lifecycle	Open, Acknowledged, Snoozed, Assigned, Resolved, Dismissed, Auto-recovered	Only permitted transitions; dismiss requires reason for critical alerts.
Delivery	Pending, Sent, Delivered, Failed, Retrying, Suppressed	Per-channel status retained; quiet hours delay, not lose, eligible messages.
Deduplication	Open event key + cool-down	Same scope/type/period is grouped; higher severity may notify immediately.

14. Testing Strategy and Acceptance Test Scenarios

14.1 Test Strategy

- Unit tests cover money, recurrence, proration, tier pricing, allocation, FX, forecast, target, state transition, and permission rules.
- Feature/integration tests cover authenticated workflows, validation, database transactions, queues, schedules, imports, exports, and notification delivery.
- End-to-end tests cover the highest-value use cases on desktop and mobile breakpoints using realistic data and role combinations.
- Security tests cover authentication, authorization, tenant isolation, injection/XSS/CSRF, IDOR, rate limits, uploads, secret leakage, masking, and audit.
- Accessibility tests combine automated scanning with keyboard, focus, zoom/reflow, screen-reader spot checks, contrast, chart alternatives, and error handling.
- Performance tests use representative high-volume workspaces for dashboard, list, report, import, target evaluation, and queue backlog behaviour.
- Recovery tests validate backup restoration, failed-job retry, import idempotency, connector outage, email outage, and rollback/roll-forward.
- User acceptance testing is passed only when all Must requirements and critical journeys meet agreed results with no open critical/high defects.

14.2 Functional and Calculation Acceptance Tests

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-001	UC-01 / Setup	Complete onboarding with BDT base currency and Asia/Dhaka time zone; add masked card and monthly subscription.	Dashboard reflects preferences, normalized cost, next renewal and starter target; no full card data accepted.	E2E
AT-002	UC-02 / Monthly	Create USD 20 monthly plan starting on the 15th with 10% discount and fixed tax.	Schedule, first-period proration rule, net cost, next charge and annualized view match configured rules.	Calc
AT-003	UC-02 / Annual	Create annual plan USD 120 renewing in six months.	Cash forecast shows renewal in correct month; normalized monthly view shows USD 10 before other components.	Calc
AT-004	Recurring dates	Create plan billed on day 31 and generate commitments across February/leap year.	End-of-month rule is applied consistently with no skipped or duplicate commitment.	Calc

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-005	UC-03 / Tier rate	Configure first 1M units included, next 4M at rate A, remainder at rate B; post 7M.	Included units and tiers apply once; total and explanation match hand calculation.	Calc
AT-006	UC-03 / Hybrid	Configure base fee with included units and overage; add usage below then above allowance.	Forecast/actual show base only below allowance and correct overage above it.	Calc
AT-007	UC-04 / Package	Buy two packages with different expiry; post consumption.	FIFO-by-expiry allocation, remaining balances and effective paid-unit cost reconcile.	Calc
AT-008	Package expiry	Let a package expire with unused units.	Balance is no longer spendable; expired quantity/cost impact remains visible and alert resolves appropriately.	Func
AT-009	UC-05 / Plan change	Future-date price increase and payment account change.	Old schedule/history retained; new commitments and account apply from effective date only.	E2E
AT-010	Version overlap	Attempt overlapping effective plan/rate ranges.	Activation is blocked with field-level explanation; no partial version is created.	Neg
AT-011	UC-06 / Reconcile	Post actual charge matching one expected commitment.	Expected item is matched/offset exactly once; actual and forecast totals do not double count.	E2E
AT-012	Duplicate charge	Import same provider reference twice.	Second row is skipped/flagged according to decision; ledger remains idempotent.	Neg
AT-013	Refund	Post refund linked to original charge.	Net cost, target utilization and audit chain update; original remains visible.	Func
AT-014	UC-13 / Correction	Correct a posted cost by reversal and replacement.	No overwrite occurs; chain balances and reports use corrected net result.	E2E
AT-015	FX history	Post foreign charge with stored rate; later update current FX rate.	Historical base amount remains unchanged; new forecasts use the approved effective rate.	Calc

14.3 Target, Alert, Renewal and Payment Acceptance Tests

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-016	UC-07 / Target	Create monthly category target with warning 80% and critical 100% using forecast basis.	Utilization and preview are correct; invalid reversed thresholds are rejected.	E2E
AT-017	UC-08 / Warning	Post charge causing forecast to cross 80%.	One warning alert is created and delivered with contributors and correct action link.	E2E
AT-018	Alert dedupe	Recalculate same unchanged warning multiple times within cool-down.	No duplicate user notification; evaluation/delivery trace remains inspectable.	Func
AT-019	Alert escalation	Increase cost from warning to critical during cool-down.	Existing event escalates and critical notification is sent immediately per policy.	Func
AT-020	Alert recovery	Reverse erroneous cost so utilization drops below threshold.	Event records auto-recovery/updated state but prior warning history is retained.	Func
AT-021	Quiet hours	Trigger non-critical alert during quiet hours.	In-app event exists; configured outbound delivery waits and sends after quiet period.	Func
AT-022	Delivery failure	Simulate email provider failure.	Delivery retries with backoff, records failure, leaves in-app alert open, and surfaces operations issue.	Integr
AT-023	Renewal reminders	Active auto-renew plan enters 30/14/7/1-day windows.	Configured reminders occur once per window in workspace time zone with correct amount and deadline.	Sched
AT-024	Trial conversion	Trial ending with automatic conversion enabled.	Reminder shows conversion price; forecast includes post-trial fee according to configured rule.	Calc

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-025	Cancellation deadline	Cancel before deadline and confirm effective/access-end dates.	Eligible future commitments stop; historical costs and evidence remain.	E2E
AT-026	Late cancellation	Record cancellation after contractual deadline.	System warns renewal may remain, requires outcome confirmation, and does not remove commitment prematurely.	Neg
AT-027	UC-10 / Replace	Bulk replace expiring card for selected subscriptions.	Future plan versions change; past entries retain old alias; excluded items are reported.	E2E
AT-028	Account expiry	Card expiry precedes next renewal.	Alert lists every affected accessible subscription and next charge without exposing sensitive data.	Func
AT-029	Payment deactivation	Deactivate account still linked to future commitments.	System blocks or requires guided reassignment/exception; impact count is shown.	Neg
AT-030	On-demand run rate	Post half-month usage and compute run-rate forecast.	Elapsed/total billable-day rule, minimum/cap and data freshness are displayed and correct.	Calc

14.4 Import, Reporting, UI and Responsive Acceptance Tests

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-031	UC-11 / Import	Upload file containing valid, warning, error and duplicate rows.	Preview counts and row messages are accurate; execution follows documented partial/atomic policy.	E2E
AT-032	Import retry	Force worker failure after batch staging and retry.	Committed rows are not duplicated; batch resumes or rolls back deterministically.	Recovery
AT-033	UC-12 / Dashboard	Apply period/category/account filters and drill from forecast KPI.	All cards/charts use same context; underlying rows reconcile to KPI and URL/state preserves filters.	E2E
AT-034	Empty state	Open new blank workspace.	No broken zero charts; onboarding guidance and permitted quick actions are shown.	UI
AT-035	Stale source	Vendor usage has not refreshed by expected time.	Dashboard/report marks stale data and explains forecast inclusion/exclusion.	UI
AT-036	Export masking	Viewer exports payment-account report.	Export matches filters and contains only permitted masked fields; event is audited.	Security
AT-037	Desktop layout	Test core pages at 1440×900.	Navigation, KPI/cards, forms and tables align; no overlap, clipping or unexplained horizontal page scroll.	Visual
AT-038	Tablet layout	Test core journeys at 768×1024 and landscape.	Drawer navigation, two-column cards and priority-table behaviour remain complete and usable.	Visual
AT-039	Mobile layout	Complete add subscription, record usage and resolve alert at 375×812.	Single-column flow, keyboard/input types, sticky actions and validation work without overlap or page-wide scrolling.	E2E
AT-040	Small mobile	Test dashboard and forms at 320 px and 200% zoom.	Core content reflows; no action is hidden; exceptions are limited to genuine tables with accessible alternative.	A11y
AT-041	Keyboard	Complete core flows without mouse.	Focus order is logical and visible; menus/modals/drawers work; focus returns correctly.	A11y
AT-042	Screen reader	Review dashboard KPIs, alert detail, form errors and chart alternative.	Names, roles, states, error associations and summaries are understandable.	A11y

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-043	Color independence	Inspect target/alert states under grayscale and color-vision simulation.	Severity remains clear through labels, icons and text; contrast passes target.	A11y
AT-044	Validation persistence	Submit invalid conditional form and receive server errors.	Specific errors appear at fields and summary; valid inputs remain; focus moves to summary/first error.	UI
AT-045	Saved view	Save filters/columns/density, reload on mobile and desktop.	Allowed preferences persist; incompatible mobile columns degrade cleanly and can be reset.	Func

14.5 Security, Performance and Operational Acceptance Tests

Test ID	Use case / area	Scenario and steps	Expected result	Type
AT-046	Tenant isolation	Attempt direct-object and crafted-query access to another workspace across web, API, export and job paths.	Every attempt is denied without existence leakage; security event is logged appropriately.	Security
AT-047	Role matrix	Exercise create/edit/correct/export/admin actions as each role.	Server-side permission outcomes match approved matrix; hidden buttons are not the only control.	Security
AT-048	Sensitive input	Attempt to enter full card number, CVV or password in payment fields/notes.	Dedicated fields never request them; prohibited data is rejected/warned and not logged.	Security
AT-049	Upload safety	Upload oversized, mismatched MIME, executable and infected test files.	Unsafe files are rejected/quarantined; no public or executable access is created.	Security
AT-050	Session security	Test reset token replay, session fixation, inactivity, logout all devices and rate limiting.	Tokens are single-use/expiring; sessions rotate/revoke; abuse receives safe responses.	Security
AT-051	Audit integrity	Change plan, correct cost, change role, export data and resolve alert.	Complete append-only audit events include actor, outcome, entity, correlation and required reason.	Audit
AT-052	Dashboard performance	Load representative high-volume workspace under agreed concurrent traffic.	95th percentile meets target; totals are correct; cache does not cross workspace/filter boundaries.	Perf
AT-053	Queue backlog	Create high alert/import workload and restart workers.	Jobs are idempotent, recover, expose progress/failure and do not lose/duplicate financial events.	Recovery
AT-054	Backup restore	Restore latest approved backup into isolated environment and run reconciliation checks.	Restore meets RPO/RTO baseline and source ledger totals/snapshots reconcile.	Recovery
AT-055	Scheduler/time zone	Run renewal and target jobs across DST/time-zone boundary and month-end.	Events use workspace commercial dates, execute once, and do not drift with server time.	Sched
AT-056	Connector outage	Simulate timeout, partial payload and duplicate replay.	Timeout/circuit/retry rules apply; data is idempotent; stale status and manual fallback are visible.	Integr
AT-057	Metronic hygiene	Inspect production bundle/routes/assets.	Unused demos/plugins/credentials are absent; dependency/license inventory is documented.	Security
AT-058	Error privacy	Trigger validation, 404/403/500 and failed background jobs.	User receives safe correlation code; logs contain diagnostic context without secrets or prohibited data.	Security
AT-059	Account closure	Export then close account with retention hold.	Sessions revoke, notifications stop, retained classes are explained, purge workflow respects hold.	Privacy
AT-060	Period close	Close month, attempt ordinary edit, then perform authorized reopen with reason.	Edit is blocked while closed; reopen and affected recalculation are fully audited.	Func

14.6 Release Acceptance Gates

- All Must requirements are implemented or have an approved, documented waiver from the Product Owner and Sponsor.
- All UC-01 through UC-15 primary flows pass; no Critical/High unresolved defect affects security, money, permission, data loss, renewal, or alert accuracy.
- Cost/forecast reference calculations reconcile at defined decimal precision and rounding tolerance.
- Responsive and accessibility acceptance passes for the supported browser/device matrix and core workflows.
- Security review, dependency scan, backup restore, migration rehearsal, monitoring, support runbooks, and production-readiness checklist are complete.
- Business owner signs off dashboard totals, target/alert behaviour, imports, exports, privacy handling, and notification templates using representative data.

15. Requirements Traceability

Capability	Objectives	Requirements	Use cases	Acceptance tests
Identity & setup	OBJ-01, OBJ-06	FR-001-006, FR-083	UC-01, UC-15	AT-001, AT-034, AT-050, AT-059
Catalogue & types	OBJ-01	FR-010-015, FR-080	UC-02, UC-03, UC-14	AT-002-005, AT-010
Recurring lifecycle	OBJ-01, OBJ-03	FR-020-028	UC-02, UC-05, UC-09	AT-002-004, AT-009-010, AT-023-026
Cost & reconciliation	OBJ-02, OBJ-04	FR-030-037	UC-06, UC-11, UC-13	AT-011-015, AT-031-032, AT-060
Token & on-demand	OBJ-02, OBJ-05	FR-040-048	UC-03, UC-04	AT-005-008, AT-030
Payment accounts	OBJ-04	FR-050-055	UC-02, UC-10	AT-001, AT-027-029, AT-036, AT-048
Targets & alerts	OBJ-02, OBJ-03	FR-060-068	UC-07, UC-08	AT-016-022
Dashboard & reports	OBJ-02, OBJ-05, OBJ-06	FR-070-077	UC-12	AT-033-045, AT-052
Admin & operations	OBJ-01-06	FR-080-087	UC-11, UC-14, UC-15	AT-031-032, AT-046-060
Security & privacy	OBJ-04	SEC-01-12, NFR-06-15	All	AT-036, AT-046-059

16. Delivery Phasing, Migration and Rollout

16.1 Recommended Release Plan

Phase	Scope	Exit criteria
Phase 0 — Foundation	Architecture, identity, workspace boundary, design system, reference data, CI/CD, logging, security baseline.	Technical runway, security model and UI component baseline approved.
Phase 1 — Core MVP	Tool/vendor/type, recurring subscriptions, masked payment accounts, ledger, dashboard, monthly/annual normalization, targets, in-app/email alerts, audit.	Must-have recurring-cost journey and core acceptance tests pass.
Phase 2 — Variable Cost	Token packages, meters/rates, usage import, on-demand/hybrid forecasting, balances, stale-data and overage alerts.	Reference pricing/usage calculations and token tests pass.
Phase 3 — Control & Insight	Reconciliation, period close, plan change/cancellation evidence, advanced reports, saved views, import/export, payment reassignment.	End-to-end monthly close and renewal-control UAT passes.
Phase 4 — Integrations & Optimization	Approved vendor/FX/connectors, scheduled reports, recommendations, optional team/workspace expansion.	Connector resilience, privacy authorization and performance gates pass.

16.2 MVP Boundary

MVP includes all Must requirements except an explicitly approved connector-dependent item that has a complete manual/import fallback. It must support personal production use with recurring, prepaid, on-demand and hybrid subscriptions; payment aliases; accurate cost ledger; targets; alerts; dashboard; responsive UI; exports; audit; security; and operational recovery.

16.3 Migration and Initial Data Load

1. Inventory source files, columns, currencies, date conventions, duplicates, sensitive data and ownership.
2. Map vendors, tools, categories, subscription types, statuses, payment aliases, cost history and usage.
3. Clean prohibited secrets/full payment numbers before files enter the application migration process.
4. Run dry-run import with row-level validation, duplicate rules, currency conversion and sample reconciliation.
5. Obtain business approval of counts, active status, next renewals, normalized costs, payment links and opening targets.
6. Execute production import in controlled window, preserve batch evidence, and reconcile dashboard/report totals.
7. Retain source files only under the approved secure retention policy; remove staging copies after acceptance.

16.4 Rollout and Change Management

- Pilot with representative monthly, annual, token, on-demand, hybrid, multi-currency, and expiring-payment scenarios.
- Provide a quick-start guide, calculation glossary, import template, notification guide, and renewal/cancellation checklist.

- Use a brief parallel reconciliation period against source statements/spreadsheets before declaring the platform authoritative.
- Measure onboarding completion, unmatched entries, stale usage, alert action rate, and user-reported calculation questions.
- Hold a post-launch review after the first complete reporting cycle and prioritize corrections before optional integrations.

17. Risks, Dependencies, Decisions and Sign-off

17.1 Risk Register

ID	Risk	Impact	Mitigation / control	Owner
R-01	Incorrect cost, FX, tier or forecast calculation	High	Central tested domain services, reference examples, explainable totals, reconciliation and UAT.	Tech Lead / QA
R-02	Users enter incomplete or stale usage/cost data	High	Freshness indicators, reminders, imports/connectors, exception queues and disclosed forecast basis.	Product Owner
R-03	Sensitive payment or credential data is collected	Critical	Data minimization, prohibited fields, input warnings, masking, secure secrets for approved connectors only.	Security
R-04	Alert fatigue causes important events to be ignored	High	Severity, dedupe, digest, quiet hours, escalation, acknowledgement analytics and sane defaults.	Product Owner
R-05	Vendor API changes or outages	Medium	Adapter isolation, monitoring, retries, circuit breaker, stale status and manual/import fallback.	Tech Lead
R-06	Metronic customization creates upgrade/accessibility debt	Medium	Application wrapper layer, token mapping, remove demos, accessibility tests and license/dependency inventory.	UX / Tech Lead
R-07	Large ledger/usage volume slows dashboards	Medium	Indexes, query tests, async summaries, caching with tenant keys, performance gates and archive policy.	DB / Tech Lead
R-08	Accidental deletion or historical overwrite	High	Soft delete, reversal workflow, plan versions, confirmation/impact preview, audit and backup restore drills.	Tech Lead
R-09	Ambiguous ownership of targets and renewals	Medium	Mandatory owner/assignee, role matrix, calendar, escalation and overdue workflow.	Product Owner
R-10	Multi-currency totals are misunderstood	Medium	Original/base display, stored rate/source/date, freshness warnings and calculation explanation.	Product Owner

17.2 External and Project Dependencies

- Confirmed business owner, prioritization authority, UAT representatives, and sample/source data.
- Valid Metronic license and approved component/plugin list.
- Hosting, TLS, database, queue/cache, email, object storage, monitoring, backup, and secret-management services.
- Decisions on authentication/MFA, tenant model, notification provider, FX source, data region, retention, and privacy terms.
- Approved calculation examples for representative vendors and on-demand/token rate structures.

17.3 Open Decisions Required Before Build Completion

Decision ID	Decision	Recommended baseline	Needed by
D-01	Single-user versus multi-user workspace at launch	Design tenant-ready; enable owner plus optional viewer/editor roles.	Architecture sign-off
D-02	Base currency and FX source	User base currency with stored dated manual/approved provider rates.	Cost module build
D-03	Tax display and recognition	Store tax components; offer inclusive/exclusive reporting; no tax filing.	Ledger design
D-04	Forecast default for on-demand	Run-rate when data is fresh; manual/target fallback with visible method.	Forecast build
D-05	Notification channels	In-app and email for launch; add channels only after separate approval.	Alert build
D-06	Financial-period close	Optional monthly close with authorized reopen and adjustment workflow.	Reporting build
D-07	Retention and hosting region	Confirm by jurisdiction and service policy before production.	Production readiness
D-08	MFA/step-up actions	Offer MFA; require step-up for role changes, full export and account closure if feasible.	Security sign-off

17.4 Acceptance and Sign-off

Approval of this BRD confirms the business problem, scope, requirements, rules, use cases, UI expectations, data/security baseline, test approach, and recommended delivery plan. Approval does not remove the need for detailed solution design, backlog refinement, data-protection/legal review, calculation examples, interface specifications, and sprint-level acceptance criteria.

Role	Name	Approval / conditions	Signature	Date
Business Sponsor				
Product Owner				
Technical Lead				
QA / Security Lead				

Appendix A — Role and Permission Matrix

Capability	Owner	Admin	Editor	Viewer / Auditor	Finance Reviewer
View dashboard/reports	Yes	Yes	Yes	Yes	Yes
Create/edit tools/subscriptions	Yes	Yes	Yes	No	Read
Change/cancel plan	Yes	Yes	Yes*	No	Read
Post ordinary cost/usage	Yes	Yes	Yes	No	Yes
Correct posted/closed cost	Yes*	Yes	No	No	Yes
Manage payment aliases	Yes	Yes*	Yes*	Masked read	Masked read
Manage targets/alerts	Yes	Yes	Yes	Read	Read
Manage users/roles/reference data	Owner	Yes	No	No	No
Export data	Yes	Yes	Scoped	Scoped	Scoped
Close account / purge	Yes	No	No	No	No
View audit	Yes	Yes	Scoped	Yes	Yes

* Permission is subject to workspace policy and may require step-up verification, impact preview, reason, or approval. The final matrix must be confirmed during solution design.

Appendix B — Lifecycle Status Models

Object	Statuses	Key transition controls
Subscription	Draft, Trial, Active, Paused, Pending Cancellation, Cancelled, Expired, Archived	Effective dates, commercial confirmation, owner permission, reason and history.
Plan version	Draft, Scheduled, Effective, Superseded, Cancelled	No invalid overlap; immutable once historically used except correction workflow.
Cost entry	Draft, Posted, Reconciled, Voided, Reversed, Refunded	Posted entries corrected by linked reversal/replacement; closed periods restricted.
Usage entry	Staged, Validated, Posted, Rejected, Superseded	Provider/batch idempotency; allocation occurs only for eligible posted usage.
Payment account	Active, Expiring, Inactive, Replaced, Archived	Impact/reassignment required before future obligations use inactive account.
Target	Draft, Active, Paused, Completed, Archived	Evaluation only when active and period/scope valid.
Alert	Open, Acknowledged, Snoozed, Assigned, Resolved, Dismissed, Auto-recovered	Severity/state history retained; critical dismissal requires reason.
Import batch	Uploaded, Staged, Validated, Processing, Completed, Partial, Failed, Rolled Back	Idempotency, counts and row outcomes retained.

Appendix C — Sample Cost Calculations

C.1 Annual Subscription Normalization

Annual gross fee USD 144; discount USD 24; non-refundable tax USD 6. Net cash cost = $144 - 24 + 6 = \text{USD } 126$. Normalized monthly management cost = $126 / 12 = \text{USD } 10.50$. The cash view posts USD 126 at renewal; the normalized view shows USD 10.50 per month.

C.2 On-Demand Tier Example

Usage is 7,000,000 units. First 1,000,000 are included, next 4,000,000 cost USD 0.50 per million, and the remainder costs USD 0.40 per million. Cost = $(4 \times 0.50) + (2 \times 0.40) = \text{USD } 2.80$, plus any effective fixed fee/tax. The calculation explanation must show included and paid quantities separately.

C.3 Target Forecast Example

Monthly target USD 100; actual through day 15 of a 30-day period is USD 45; remaining committed cost is USD 30. Run-rate variable forecast is $\text{USD } 45 / 15 \times 30 = \text{USD } 90$ only when the entire actual is treated as variable. If USD 30 of actual is recurring and USD 15 is variable, the preferred component forecast is USD 30 recurring actual + $(\text{USD } 15 / 15 \times 30)$ variable + USD 30 remaining commitments = USD 90. The UI must state which method/components were used.

C.4 Multi-Currency Entry

A charge of EUR 20 is posted using stored rate 1 EUR = 1.10 USD. Base amount is USD 22.00. Later FX changes do not rewrite the posted USD 22.00. A future commitment may use a newer approved rate and must show its forecast nature and rate date.

Appendix D — Glossary and Abbreviations

Term	Meaning
Actual	Eligible posted net cost for the selected period.
Committed	Known future obligation expected in the selected period.
Forecast	Projected final period cost based on actuals, commitments and variable-cost assumptions.
MTD / EOM	Month to date / end of month.
FX	Foreign exchange conversion between original and reporting currency.
FIFO-by-expiry	Consume the eligible balance that expires earliest first.
MoSCoW	Must, Should, Could, Won't/Out-of-scope prioritization.
RPO / RTO	Recovery point objective / recovery time objective.
WCAG	Web Content Accessibility Guidelines.
PAN / CVV	Card account number / card verification value; prohibited from storage in this product.
Idempotent	Safe to retry without creating duplicate business effects.
Metronic	Licensed UI template/component package used as the presentation foundation.

Appendix E — Definition of Ready and Definition of Done

Definition of Ready

- Requirement/use case and priority are identified; business rule and calculation examples are clear.
- UX state, role/permission, validation, data fields, audit, notification and error behaviour are specified.
- Dependencies, migration/integration needs, security/privacy considerations and acceptance tests are understood.

Definition of Done

- Code, migration, authorization, validation, audit, responsive UI and accessibility behaviour are implemented and reviewed.
- Automated unit/feature/E2E tests pass; performance/security checks appropriate to risk are complete.
- Documentation, telemetry, operational runbook, migration/rollback and user guidance are updated.
- Product Owner accepts the requirement against representative data with no disqualifying defect.